*** PYTHON-MYSQL ***

```
# pip i mysql-connector-python-----------
import mysql.connector as ms

# Create Database Connection-----------

mydb = ms.connect(
    host="localhost",
    user="root",
    password="",
    database = "ajaypy"  # Apply All Query in Database
)

mycursor = mydb.cursor()

# Create database----------------------------------------
mycursor.execute("CREATE DATABASE ajaypy")
print("Database created!")

# Show database-----------------------------------------
mycursor.execute("SHOW DATABASES")
for x in mycursor:
    print(x)



# --------Create Table -----------------------------------------
mycursor.execute("CREATE TABLE customers (id INT AUTO_INCREMENT PRIMARY
KEY, name VARCHAR(255), address VARCHAR(255))")

# ---------Alter Table -----------------------------------------
mycursor.execute("ALTER TABLE customers ADD COLUMN age VARCHAR(255)")

# ----------Insert single Data
-----------------------------------------
sql = "INSERT INTO customers (name, address, age) VALUES (%s, %s, %s)"
val = ('Peter', 'Lowstreet 4',34)
```

```
mycursor.execute(sql, val)
mydb.commit()
print(mycursor.rowcount, "record inserted.")

# ----------Insert Multiple Data
------------------------------------------
val = [
('Peter', 'Lowstreet 4',34),
('Amy', 'Apple st 652',20),
('Hannah', 'Mountain 21',21),
]
mycursor.executemany(sql, val)
mydb.commit()
print(mycursor.rowcount, "record inserted.")

# -----------Show Last Row Id
-----------------------------------------------------
print("1 record inserted, ID:", mycursor.lastrowid)


# ---------Select All
Data-----------------------------------------------------
mycursor.execute("SELECT * FROM customers")
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# ----------Select Name Column
Data-----------------------------------------------------
mycursor.execute("SELECT name FROM customers")
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

#------------- Single Row Data
-------------------------------------------------
myresult = mycursor.fetchone()
print(myresult)

# --------------Where ------------------------------------------------
```

```
sql = "SELECT * FROM customers WHERE address='Lowstreet 4'"
mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)
#-------------------
sql = "SELECT * FROM customers WHERE address = %s"
adr = ('Highway 21',)
mycursor.execute(sql,adr)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# Like ----------------------------------------------------
sql = "SELECT * FROM customers WHERE address Like '%Low%'"
mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# Order By & DESC-----------------------------------------------------
sql = "SELECT * FROM customers ORDER BY name "
sql = "SELECT * FROM customers ORDER BY name DESC"
mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# Delete Row
Data-----------------------------------------------------------
sql = "DELETE FROM customers WHERE address = 'Mountain 21'"
mycursor.execute(sql)
#--------------------
sql = "DELETE FROM customers WHERE address = %s"
adr = ("Lowstreet 4", )
mycursor.execute(sql,adr)
mydb.commit()
print(mycursor.rowcount, "records deleted")

# Drop Table ----------------------------------------------------------
```

```
sql = "DROP TABLE customers"
mycursor.execute(sql)
#-------------------
sql = "DROP TABLE IF EXISTS customers"
mycursor.execute(sql)

#---------------- Update Table Data
-----------------------------------------------------
sql = "UPDATE customers SET address = 'Mumbai 105' WHERE address = 'Apple
st 652'"
mycursor.execute(sql)
mydb.commit()
print(mycursor.rowcount, "records affected")
#----------------------
sql = "UPDATE customers SET address = %s WHERE address = %s"
val = ("Apple st 652","Mumbai 105",)
mycursor.execute(sql,val)
mydb.commit()
print(mycursor.rowcount, "records affected")

#--------------- Limit & Offset
-----------------------------------------------------------
mycursor.execute("SELECT * FROM customers LIMIT 4")
mycursor.execute("SELECT * FROM customers LIMIT 3 offset 2")
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# ----------------Create Products
Table-------------------------------------------------------------
mycursor.execute("CREATE TABLE products (id INT AUTO_INCREMENT PRIMARY
KEY,  name VARCHAR(255))")
#---------------
mycursor.execute("ALTER TABLE customers ADD COLUMN fav VARCHAR(255)")
#---------------
sql = "INSERT INTO products (name) VALUES (%s)"
val = [('Laptop',), ('Mobile',),('Laptop',),("Keyboard",)]
mycursor.executemany(sql, val)
mydb.commit()
print(mycursor.rowcount, "record inserted.")
```

```
#----------------
sql = "INSERT INTO customers (name, address, age,fav) VALUES (%s, %s, %s,
%s)"
val = ('Peter', 'Lowstreet 4',34,3)
mycursor.execute(sql, val)


# Inner Join
-----------------------------------------------------------------
sql = "SELECT " \
"customers.name AS customer," \
"products.name AS favourite  " \
"FROM customers  " \
"INNER JOIN products ON customers.fav = products.id "

mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# Left Join
-----------------------------------------------------------------------
sql = "SELECT " \
"customers.name AS customer," \
"products.name AS favourite  " \
"FROM customers  " \
"LEFT JOIN products ON customers.fav = products.id "

mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)

# Right Join
-----------------------------------------------------------------------
sql = """ SELECT
customers.name AS customer,
products.name AS favourite
FROM customers
RIGHT JOIN products ON customers.fav = products.id """
```

```
mycursor.execute(sql)
myresult = mycursor.fetchall()
for x in myresult:
    print(x)
```